

WASP-10b

R-1928

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_150821 · created 2026-08-06T01:27:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457255.7492 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.209 +/- 0.059
Transit depth (Rp/Rs)^2	4.37 +/- 2.48 [%]
Orbital Inclination (inc)	89.27 +/- 1.87
Airmass coefficient 1 (a1)	1.07 +/- 0.85
Airmass coefficient 2 (a2)	0.03 +/- 0.9
Scatter in the residuals of the lightcurve fit is	95.86 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	3.0
Transit Duration (day)	0.093 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.209 ± 0.059 vs 0.1592 (deviation 0.77σ)
Duration fit vs published	0.093 d vs 0.0946 d (deviation 0.13σ)
Mid-time O–C	7.1 min
Depth SNR	3.2
Residual scatter	95.86 %

Light curve

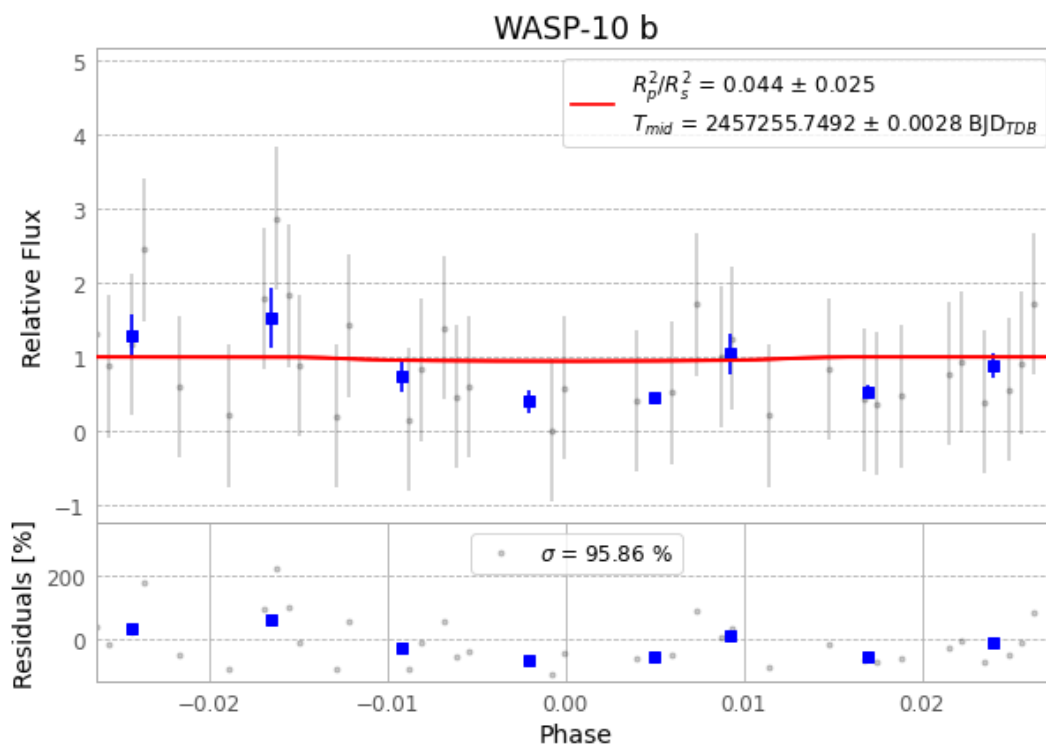


Figure 1 — FinalLightCurve_WASP-10 b_2015-08-20.png (SHA-256 874047aa7252b25b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [254, 442], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 41d8fffb83404cf9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), WASP-10150821035412.FITS → WASP-10150821075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-150821.log (1c7fdb606162...), FinalLightCurve_WASP-10 b_2015-08-20.png (874047aa7252...), FinalLightCurve_WASP-10 b_2015-08-20.csv (a065e419f706...)

Compute resources

Wall time 193.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 01:27Z.